Assignment4: Image Compression

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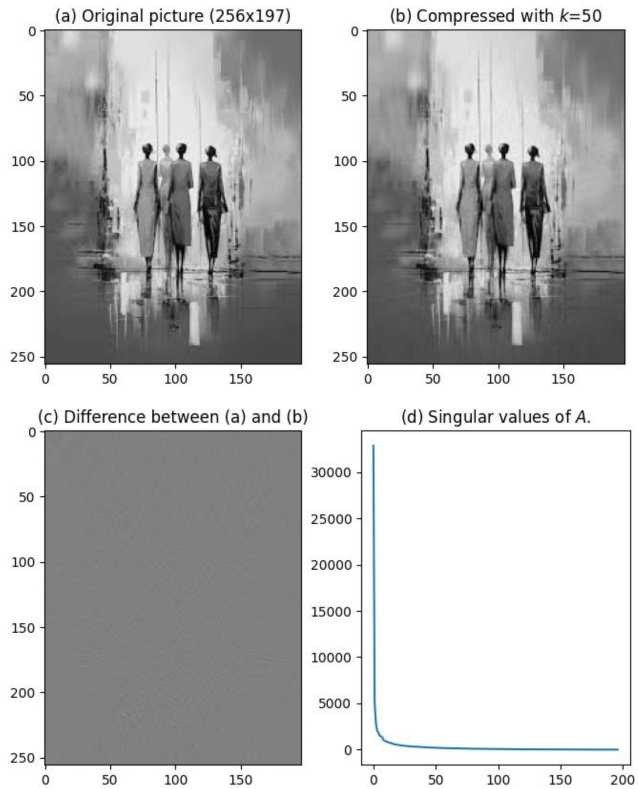
Question 2) used numbers found using code and then inserting into CHATGPT to calcualte



The size of the image is: 256 x 197

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Compression ratio is: 0.450111

Average relative difference is: 0.031967

For ρ<1:

- k=1
- Compression Ratio (ρ) = 0.009
- Average Relative Difference (δ) = 0.99

For ρ<0.5:

- k=1
- Compression Ratio (ρ) = 0.009
- Average Relative Difference (δ) = 0.99

For δ<5%:

- Compression Ratio (ρ) = 1.386

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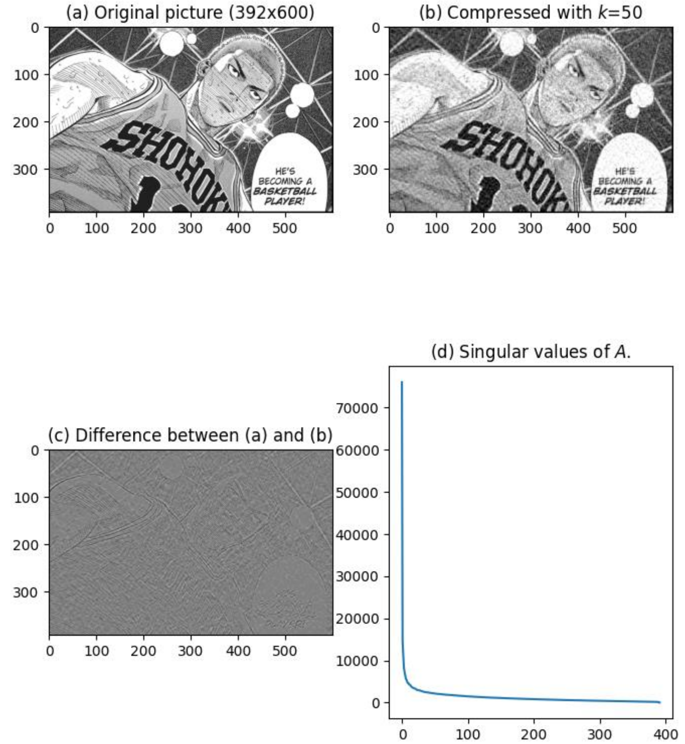
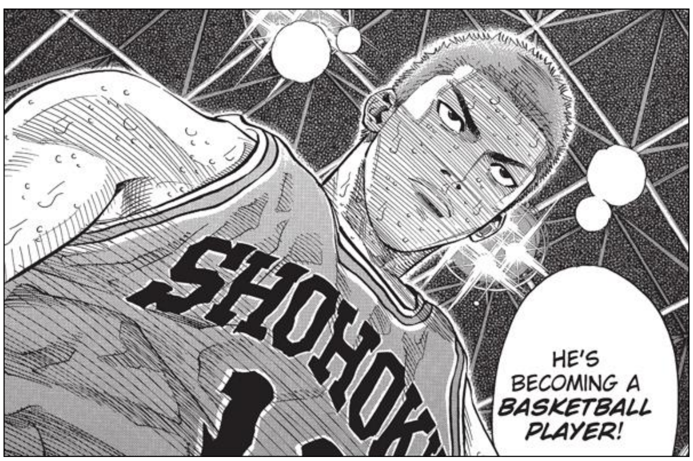




The size of the image is: 392 x 600

Compression ratio is: 0.211097

Average relative difference is: 0.365248

For ρ<1:

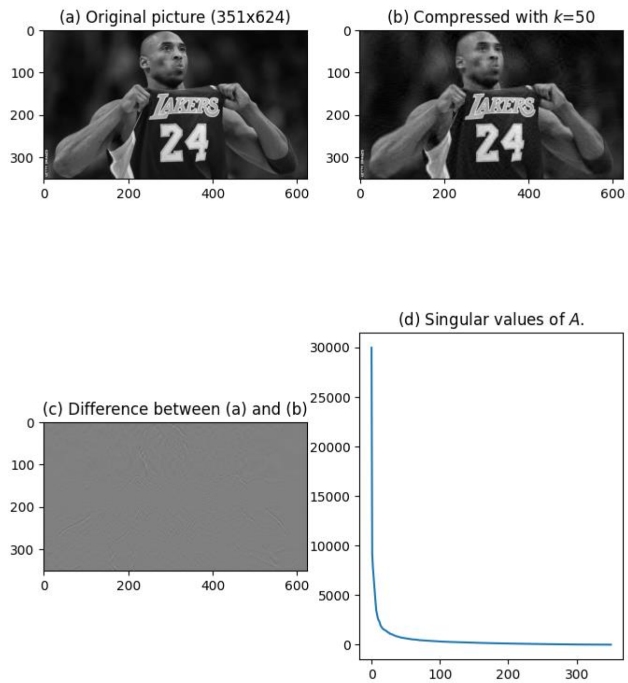
- k=359
- Compression Ratio (ρ) = 1
- Average Relative Difference (δ) = 0.224

For ρ<0.5:

- k=180
- Compression Ratio (ρ) = 0.5
- Average Relative Difference (δ) = 0.542

For δ<5%:

- k=530
- Compression Ratio (ρ) = 1.48



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The size of the image is: 351 x 624

Compression ratio is: 0.222807

Average relative difference is: 0.146808

For ρ<1:

- k=225
- Compression Ratio (ρ) = 1
- Average Relative Difference (δ) = 0.130

For ρ<0.5:

- k=113
- Compression Ratio (ρ) = 0.5
- Average Relative Difference (δ) = 0.461

For δ<5%:

- k=274
- Compression Ratio (ρ) = 1.22

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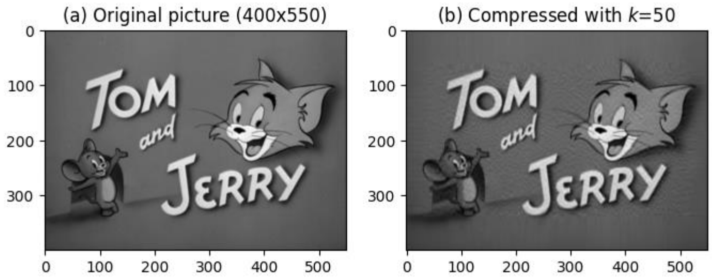

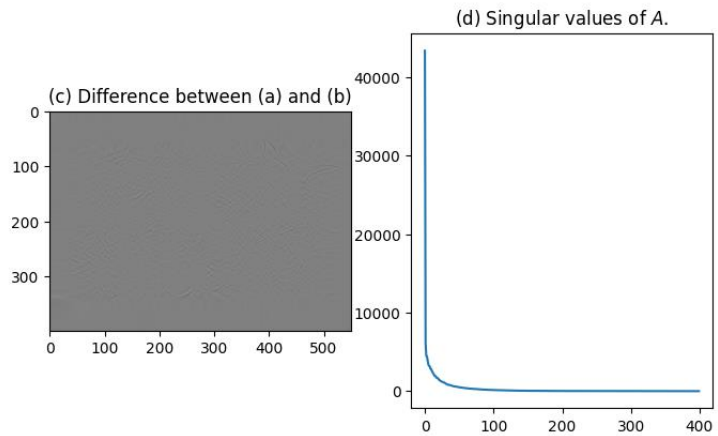

The size of the image is: 400 x 550

Compression ratio is: 0.216136

Average relative difference is: 0.047659

For ρ<1:

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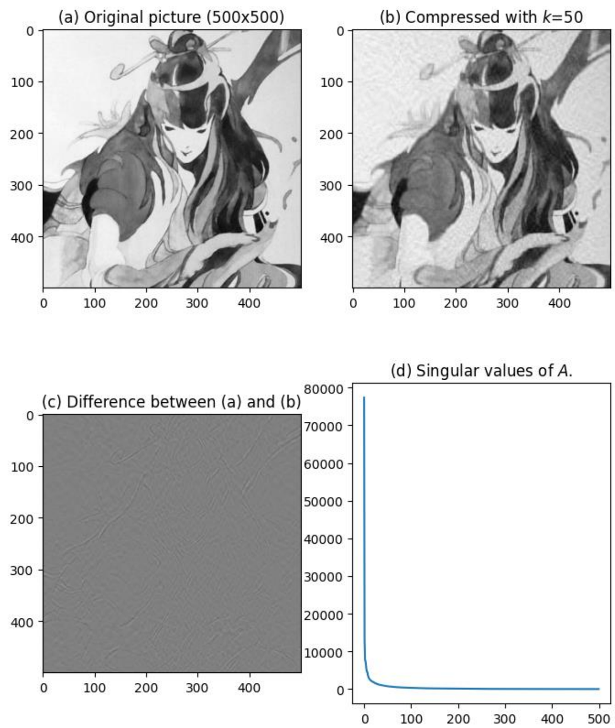
- k=232
- Compression Ratio (ρ) = 1
- Average Relative Difference (δ) = 0.177

For ρ<0.5:

- k=116
- Compression Ratio (ρ) = 0.5
- Average Relative Difference (δ) = 0.505

For δ<5%:

- k=311
- Compression Ratio (ρ) = 1.344



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The size of the image is: 500 x 500

Compression ratio is: 0.200200

Average relative difference is: 0.051343

For ρ<1:

- k=250
- Compression Ratio (ρ) = 1
- Average Relative Difference (δ) = 0.250

For ρ<0.5:

- k=125
- Compression Ratio (ρ) = 0.5
- Average Relative Difference (δ) = 0.563

For δ<5%:

- k=389
- Compression Ratio (ρ) = 1.558

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Question 3a) CHATGPT and some online research

Pictures that are easily compressible by SVD (resulting in smaller relative differences) typically have these characteristics:

1. **Simple Patterns**: Uniform colors, smooth gradients, or structured geometric designs (e.g., logos, cartoons).
2. **High Redundancy**: Repetitive or symmetric patterns.
3. **Low Detail**: Few edges, minimal fine details, or smooth transitions (e.g., plain backgrounds, sunsets).
4. **Dominant Features**: One main object with a plain background.
5. **Grayscale Images**: Simpler compared to RGB images due to fewer dimensions.

Hard-to-Compress Images:

- High-detail, noisy, or highly textured images (e.g., forests or crowds).
- Sharp transitions or abrupt changes (e.g., text-heavy images).

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Question 3b) CHATGPT

1. Singular Value Distribution:

- **Sharp Decay in Singular Values:**
  - For easily compressible images, the singular values of the matrix representing the image decay rapidly.
  - Most of the "energy" (information) is concentrated in the top singular values, with the rest being relatively small.
  - This allows the image to be reconstructed accurately using only a small number of singular values (k), leading to lower ρ and δ.

2. Low ρ(Compression Ratio):

- A low ρ indicates that the image can be represented with a small fraction of its singular values compared to the total.
- **Reason:** Rapidly decaying singular values mean fewer components (smaller k) are needed to capture the majority of the image's content.

3. Low δ (Relative Difference):

- A low δ suggests that truncating smaller singular values has a minimal impact on the reconstruction.
- **Reason:** When singular values decay sharply, the smaller singular values contribute less to the overall structure of the image, so discarding them introduces less error.

4. Hard-to-Compress Images:

- For images with slowly decaying singular values:
  - More singular values are required to preserve quality, leading to higher ρ and δ.
  - These images distribute their "energy" more evenly across many singular values, making truncation less effective.

Conclusion:

The ease of SVD compression is directly tied to the rate of decay of the singular values:

- **Fast decay:** Fewer singular values retain most of the image's content, resulting in low ρ and δ.
- **Slow decay:** Singular values are more evenly distributed, requiring more components to preserve quality, leading to higher ρ and δ.

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Question 3c) CHATGPT

The largest differences in compressed images occur in:

1. **High-Frequency Areas (Edges and Fine Details):**
   - Sharp edges and intricate details are lost because smaller singular values, which encode fine details, are truncated.

2. **High-Contrast Regions:**
   - Areas with abrupt pixel value changes (e.g., between light and dark) suffer from smoothing or blurring due to the loss of higher singular values.

3. **Gradients and Shadows:**
   - Smooth transitions may develop **banding artifacts** (visible steps) as they are approximated with fewer singular values.

Why These Areas Differ:

SVD prioritizes global structures by retaining the most significant singular values. Smaller singular values, which encode localized details, are removed during compression, leading to visible differences in detailed or high-contrast regions.